KAUSTUBH DUA

Software Developer | Electronics & Computer Engineering, JIIT Noida | Class of 2029

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SUMMARY

Student developer at the intersection of **AI, data, systems and product engineering**. I turn messy, real workflows (crop residue, waste routing, multilingual help, security triage) into tested tools with clear system boundaries, honest status labels and verifiable claims. Working style: **build thoughtfully, verify everything**.

EDUCATION

B.Tech., Electronics & Computer Engineering (ECM) — Expected graduation **2029**
Jaypee Institute of Information Technology (JIIT), Noida

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++17, SQL
Application: React, Next.js, Vite, FastAPI, Streamlit, Tailwind CSS
Data, AI & optimisation: PostgreSQL / PostGIS, Supabase, pandas, scikit-learn, Google OR-Tools, Ollama (local inference), embeddings, Three.js / WebGL
Quality & delivery: Git, GitHub Actions, pytest, Playwright, CodeQL, Docker, CTest

SELECTED PROJECTS

NIRDHOOM | React 19, Supabase, PostGIS, OR-Tools, Leaflet
- Crop-residue coordination system linking field requests, service workflows, operational evidence and buyer pathways.
- Service keys stay on the server, never in the browser; row-level security guards the data.

GitHub · Live

Saathi | Next.js, FastAPI, speech APIs
- Voice-first, multilingual assistant built around source-aware answers, consent-based workflows and human escalation.
- Ships a 120-case synthetic benchmark for its Hindi/Hinglish intent router; labels demo and live results.

GitHub · Live

SATARK | Python, Streamlit, LLM / vision workflows
- AI-assisted security-awareness workspace for triaging suspicious messages, links, images, QR codes, PDFs and supported videos.

GitHub · Live

PatchPilot | Python, FastAPI, pytest, Docker
- Explainable pull-request risk triage that never clones or executes the code it analyses.

GitHub

Smart Waste Collection | Python, Random Forest, OR-Tools, Streamlit
- Predict-prioritise-route pipeline for waste collection. States plainly that its IoT-style data is synthetic.

GitHub · Live

GitGlobe | React Three Fiber, WebGL, FastAPI, embeddings
- Interactive 3D map for exploring open-source repositories by capability.

GitHub · Live

COMMUNITY & TEAM WORK

- **JIIT Youth Club website** (React 19, Vite, Supabase): public site for the club behind major events at the Sector 128 campus, guarded by GitHub Actions, CodeQL and Playwright browser checks. Live
- **Adapt AI**: contributor on a team-built local AI assistant that checks hardware and recommends a suitable on-device model (FastAPI, Ollama).
- **This portfolio**: a 3D village and Nether-fortress world built with a hand-written WebGL 2 engine (no 3D library), tested with Playwright. Source

ALSO BUILT

AkashChalak (air-quality hotspot decision support) • Smart Bike Taxi Platform (C++17 graph algorithms, CTest) • JIIT Toppers (Next.js, Prisma) • Weather Core Matrix and Nord Studio Task Manager (vanilla JS)